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CAPABILITIES IN AN (ECONOMIC) THEORY OF FIRMS

Author(s): DAVID J. TEECE

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S Y M P O S I U M

THE FOUNDATIONS OF ENTERPRISE PERFORMANCE: DYNAMIC AND ORDINARY CAPABILITIES IN AN (ECONOMIC) THEORY OF FIRMS

DAVID J. TEECE
University of California, Berkeley

The dynamic capabilities framework has had a significant impact on strategic management theory and practice, but the sizable literature on the topic has not always been unified. This paper begins with a restatement of the framework encompassing clarifications and extensions that have occurred since it was introduced. The paper highlights key elements that have been omitted or poorly integrated into the dynamic capabilities literature: the role of individual action by entrepreneurial managers, the role of resources, strategy, and the distinction between ordinary and dynamic capabilities. Dynamic capabilities is advanced as a multidisciplinary framework to explain long-run enterprise performance. Ambidexterity and other related frameworks are tailored versions of dynamic capabilities. Linkages between (strategic) management theory and (Austrian) economic theory are explored. The concepts of x-inefficiency and d-ineffectiveness are compared.

In this paper, I outline the dynamic capabilities framework in a manner that I hope will aid future research by reconciling and integrating various perspectives. In particular, I integrate the resources and capabilities paradigms in the field of strategic management while analytically separating (and relating) strategy and capabilities. I hope that packaging dynamic capabilities concepts in this manner will help scholars and practitioners alike better understand the framework and allow it to be applied more consistently.

To address some inchoate dimensions of the dynamic capabilities literature, this paper begins with a restatement of the framework incorporating clarifications and extensions that have occurred since it was introduced.

One of the fundamental concepts in the framework is that of capability. An enterprise capability is a set of current or potential activities that utilize the firm's productive resources to make and/or de-

liver products and services. There are two important classes of capability: ordinary and dynamic. Ordinary capabilities involve the performance of administrative, operational, and governance-related functions that are (technically) necessary to accomplish tasks. Dynamic capabilities involve higher-level activities that can enable an enterprise to direct its ordinary activities toward high-payoff endeavors. This requires managing, or "orchestrating," the firm's resources to address and shape rapidly changing business environments.

The dynamic capabilities framework was created with an ambitious agenda in mind, namely to provide a general framework to help scholars and practitioners understand the foundations of firm-level competitive advantage and associated enterprise value creation and maintenance. The context assumed was business environments where there is strong innovation-driven competition, often global in scope. It was hypothesized that strong dynamic capabilities and good strategy anchored by difficult-to-imitate resources are the basis for the sustained competitive advantage displayed by a handful of firms that have endured for decades, even as they may have shifted the focus of their activities.

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The fundamental question being asked—perhaps the most fundamental question in strategic management—is “What is the inherent source of enterprise-generated future cash flows?” Figuring out the foundations or inherent sources of long-term cash flow generation is, at a deep level, one of the greatest conundrums in economic and financial theory. Despite its importance, it receives little attention outside the field of strategic management. Neither financial theory nor economic theory attempts to explain the underpinnings of cash flow generation, even though cash flow is the touchstone of enterprise valuation methodologies. The question is rarely addressed at a fundamental level, not because it is unimportant, but because it is extremely difficult to answer at any level of generality. It is tantamount to explaining the foundations of the firm’s (long-run) competitive advantage.

The field of strategic management has accepted the challenge and valiantly endeavors to uncover the fundamental sources of competitive advantage. In particular, the “resources” and “dynamic capabilities” approaches grapple with the question and have had a significant impact on mainstream management theory and practice. Yet, as shown bibliometrically by Peteraf, Di Stefano, and Verona (2013), the sizable literature that has sprung up under the dynamic capabilities rubric has split into at least two distinct streams based on different assumptions and perspectives. One purpose of this paper is to address the issues laid out by Peteraf et al. in their *Strategic Management Journal* article and in their follow-on article in this issue (Di Stefano, Peteraf, & Verona, 2014). Another goal of this paper is to more clearly explain how dynamic capabilities and strategy work together to shape competitive advantage at the level of the enterprise.

The paper begins by reviewing how some versions of the dynamic capabilities framework have (perhaps inadvertently) lost sight of the original articulation and subsequent refinements. The refinements include the distinctions between ordinary and dynamic capabilities and the inclusion of non-routine managerial action as a type of dynamic capability. The penultimate section of the paper contrasts the dynamic capabilities framework with mainstream and Austrian economic theory. With these refinements and distinctions more clearly articulated, it becomes easier to reconcile various branches of the literature. A concluding section summarizes the main points.

THE DYNAMIC CAPABILITIES FRAMEWORK: AN UPDATE AND SUMMARY¹

I begin this exploration of the dynamic capabilities framework by summarizing my current thinking on its nature. The framework is most relevant in the growing number of “ecosystems” characterized by what D’Aveni (1994) called hypercompetition, or what I call dynamic competition or “next-generation competition” (Teece, 2012a).² The framework has been deepened in ways that make linkage to other concepts in strategic management and economics more apparent. After more carefully describing how I use the terms ordinary and dynamic capabilities, I return to a discussion of the validity of the framework as an analytic lens into the activities of organizations.

A capability, ordinary or dynamic, can be harnessed, against the opposition of circumstance, to produce desirable outcomes. It is distinct from an organization’s intentions, motivations, or strategy. Capabilities are not appropriately summarized by a production function because capabilities are what the organization could accomplish, not necessarily what it is currently producing. Capabilities arise in part from learning, from organizational resources, and from organizational histories. They are untethered from particular purposes or products. For example, a capability to make machines powered by small, compact internal combustion engines can manifest itself in the manufacturing of automobiles, outboard (boat) motors, or tractors and lawn mowers.

Dynamic capabilities govern other organizational activities. They can allow an enterprise to generate superior profits by developing and producing differentiated products and services that address new and existing markets where demand is robust. They enable the firm to integrate, build, and reconfigure internal and external resources to maintain leadership in continually shifting business environments. Strong dynamic capabilities enable the firm to produce not just the best of a product type, but something that is unique and exceptional in the value it

¹ Some elements from this section can be found in Teece (2014).

² Next-generation competition is characterized by market structures that are fluid, clusters of know-how and technologies that are globally dispersed, competition that takes place between business ecosystems, and innovations that are dependent on combinations of technologies (see Teece, 2012a, Table 1).

provides to the buyer and the return it generates for the shareholder and other stakeholders.

But strong dynamic capabilities alone are unlikely to result in competitive advantage. Difficult-to-imitate (idiosyncratic) resources and good strategy are necessary, too. Thus, combined, the strength of a firm's dynamic capabilities determines the speed and degree to which the firm's idiosyncratic resources can be aligned and re-aligned consistent with the firm's strategy.

Dynamic capabilities have also been called higher-order capabilities, a phrase that appears to come from Collis (1994), who adapted the idea of higher-order derivatives from differential calculus. In Collis' taxonomy, competitive advantage is about learning how to learn and can be undermined by a rival that is better at learning how to learn how to learn, and so on, in a process that leads to infinite regress with ever higher orders of capability such that no level of capability can provide a durable advantage. Zollo and Winter (2002) focused mainly on first- and second-order capabilities but said that, in fast-changing and unpredictable contexts, "even the higher-order learning approaches will themselves need to be updated repeatedly" (p. 341), which has the flavor of infinite regress about it. These perspectives stem from a very routine-focused view of dynamic capabilities, which may be where the concern about learning to learn and infinite regress comes from. Collis (1994) mitigated the regress problem by suggesting that the relevance of a given level of higher-order capabilities is dependent on the competitive context; some industries demand faster, more flexible learning and transformation. The prospect of regress becomes greatly diminished once one introduces non-routine managerial actions (Teece, 2012b), as I discuss later.

I now turn to a closer examination of these first- and second-order concepts, which I prefer to call ordinary and dynamic capabilities. I then place these in the context of the broader dynamic capabilities framework and discuss its central claim, namely, that (some) organizations can adjust as the business environment shifts. Sometimes they may actually help shape the environment, too.

Ordinary Capabilities

An understanding of dynamic capabilities can be sharpened by comparing them with ordinary capabilities. Ordinary capabilities have also been called static (Collis, 1994), zero-level (Winter, 2003), first-

order (Danneels, 2002), and substantive (Zahra, Sapienza, & Davidsson (2006). The zero-, first-, and second- typology is used by Easterby-Smith and Prieto (2008) and Schilke (2014). The more common usage seems to be equating first-order with ordinary.

I prefer to simply distinguish between ordinary and dynamic capabilities. The ability to build new dynamic capabilities is just a part of the firm's bundle of dynamic capabilities. In my view, little is gained from a more granular typology.

Ordinary capabilities permit some degree of sufficiency (and possibly excellence) in the performance of a well-delineated task. They generally fall into three categories: administration, operations, and governance.³ Ordinary capabilities are embedded in some combination of (1) skilled personnel, including, under certain circumstances, independent contractors; (2) facilities and equipment; (3) processes and routines, including any supporting technical manuals; and (4) the administrative coordination needed to get the job done.

Ordinary capabilities can be measured against the requirements of specific tasks, such as labor productivity, inventory turns, and time to completion, and can thus be benchmarked internally or externally to industry best practices. Best operational practices are those that increase speed, quality, and efficiency. Best management practices include those that "continuously collect and analyze performance information, that set challenging and interlinked short- and long-run targets, and that reward high performers and retrain/fire low performers" (Bloom, Genakos, Sadun, & Van Reenen, 2012, p. 13). Ordinary capabilities are considered strong when the firm has achieved best practices and its employee base includes the relevant skilled people and advanced equipment.

However, best practices alone are generally insufficient to undergird sustainable competitive advantage, except in weak competitive environments. This is because much of the knowledge behind ordinary capabilities can be bought through consul-

³ In Winter's framework (e.g., Helfat & Winter, 2011), ordinary capabilities are equated with operational capabilities. Winter (2003) had earlier defined zero-level (ordinary) capabilities very strictly as doing the same thing repeatedly. In this view, any change, such as new product development or opening a new branch of a retail chain, involves a dynamic capability. I find this definition of ordinary capabilities too narrow to encompass the full range of activities involved.

tants or through a modest investment in training (Bloom et al., 2013). As a consequence, good and even “best” practices diffuse rather quickly among at least some firms in any environment open to global competition where firms have access to benchmarking data, competitive off-the-shelf technologies, and best practice training. Moreover, the management consulting industry works hard to introduce clients to the latest—and usually non-proprietary—“best” practices, allowing organizational innovations to diffuse steadily from leading firms to their rivals and to firms in other industries. This pattern of diffusion was observed, for example, with respect to the implementation of the multidivisional (M-form) organizational structure in large-scale corporations in the middle of the 20th century.⁴

In globally competitive industries, best practices are now close to universal. Bob Lutz (2011), the former vice chairman at General Motors, illustrates this point for the automotive industry:

The operations portion of the automobile business has been thoroughly optimized over many decades, doesn't vary much from one automobile company to another, and can be managed with a focus on repetitive process. It is the “hard” part of the car business and requires little in the way of creativity, vision or imagination. Almost all car companies do this very well, and there is little or no competitive advantage to be gained by “trying even harder” in procurement, manufacturing or wholesale.

More important, the presence of strong and even differentiated ordinary capabilities says nothing about whether the current production schedule is the right (or even a profitable) path to follow in the future. When the firm's output is tuned to what the market desires, strong ordinary capabilities may be sufficient for competitive advantage, but only until conditions change.

Meanwhile, best practices can become a trap, as the relentless and single-minded pursuit of efficiency can drive out the capacity to effectuate change, making the organization sclerotic. Efficiency is easiest to achieve if the tasks the organi-

zation is to perform remain fixed. Hence, inertia is often inadvertently imposed by efforts to achieve best practice. Moreover, the perceived need to maintain best practice and high productivity can distract top management from bringing about change.

In short, ordinary capabilities can best be thought of as achieving technical efficiency and “doing things right” in the core business functions of operations, administration, and governance. Even though there is considerable dispersion, a relatively high level of (basic) ordinary capabilities exists in industries open to global competition and in most firms in developed countries. Such capabilities often have a high public domain component, and, even if not, they are readily imitable and can therefore generally be acquired.

I don't mean to denigrate the importance of ordinary capabilities; they are often fundamental and can support competitive advantage for decade-long periods. Indeed, in developing countries, mastering existing technologies and practices may be more important than innovation. But on their own, they won't bring long-run success unless competition is weak because of governmentally imposed barriers to competition or other institutional and cultural barriers to competition.

Whereas ordinary capabilities are about doing things right, dynamic capabilities are about doing the right things, at the right time, based on new product (and process) development, unique managerial orchestration processes, a strong and change-oriented organizational culture, and a prescient assessment of the business environment and technological opportunities.⁵ Table 1 lays out these basic distinctions, which are developed and maintained throughout this paper.

At the risk of further oversimplifying, efficiency is at the heart of ordinary capabilities; dynamic capabilities are about adapting, orchestrating, and innovating. Strong dynamic capabilities allow

⁴ In the petroleum industry, the M-form structure diffused to a majority of the leading firms over a period of about 15 years (Armour & Teece, 1978). Once this organizational best practice became widely adopted, the econometric results show that the higher profits associated with its early adoption in the U.S. petroleum industry dissipated.

⁵ Many discussions of operations strategy drift into what I think of as dynamic capabilities. Some scholars see operations strategy as developing resources and configuring processes so that there is good strategic fit with the business environment (Van Mieghem, 2008). In the fast food industry, for example, ordinary capabilities involve key performance indicator metrics, training systems, motivation, monitoring, and so on. Dynamic capabilities address figuring out new products to put on the menu, new operating hours (e.g., late night), and new locations (central versus suburban).

TABLE 1
Some Differences Between Ordinary and Dynamic Capabilities

	Ordinary capabilities	Dynamic capabilities
Purpose	Technical efficiency in business functions	Achieving congruence with customer needs and with technological and business opportunities
Mode of attainability	Buy or build (learning)	Build (learning)
Tripartite schema	Operate, administrate, and govern	Sense, seize, and transform
Key routines	Best practices	Signature processes
Managerial emphasis	Cost control	Entrepreneurial asset orchestration and leadership
Priority	Doing things right	Doing the right things
Imitability	Relatively imitable	Inimitable
Result	Technical fitness (efficiency)	Evolutionary fitness (innovation)

firms to stay congruent with market and technological developments, as well as with broader societal goals.⁶ The late Steve Jobs made a strong statement (Kirkpatrick, 1998) with respect to the importance of spending money on the right things (i.e., effectiveness) when confronted with strong competition: “Innovation has nothing to do with how many R&D dollars you have. When Apple came up with the Mac, IBM was spending at least one hundred times more on R&D. It’s . . . about . . . how much you get it.”

While Jobs’ statement is insightful, it is also enigmatic. The dynamic capabilities framework endeavors to unlock this enigma and explain what Jobs might have meant by “get it.” Doing so requires one to go beyond organizational “fit” and a financial-statement view of enterprise strength to identify processes and managerial capacities and traits that can help the enterprise identify latent customer needs and the most promising technological opportunities, then orchestrate the resources needed to innovate, or co-innovate. In short, to understand how firms and managers “get it” requires an exploration of dynamic capabilities.

Dynamic Capabilities

Strong dynamic capabilities help enable an enterprise to profitably build and renew resources and assets that lie both within and beyond its boundaries, reconfiguring them as needed to innovate and respond to (or bring about) changes in the

market and in the business environment more generally (Pisano & Teece, 2007; Teece, Pisano, & Shuen, 1997). They allow the enterprise and its top management to develop conjectures about the evolution of consumer preferences, business problems, and technology; validate and fine-tune them; and then act on them by realigning assets and activities to enable continuous innovation and change. Successfully building strong dynamic capabilities allows firms to challenge competitors that are enamored with the resources they currently possess, that ignore (or are ignorant of) changing customer needs, that cherish the status quo, that fail to empower entrepreneurs and change agents, and that prioritize efficiency over innovation.

For applied purposes, dynamic capabilities can usefully be broken down into three primary clusters: (1) identification, development, codevelopment, and assessment of technological opportunities in relationship to customer needs (sensing); (2) mobilization of resources to address needs and opportunities, and to capture value from doing so (seizing); and (3) continued renewal (transforming). Engagement in continuous or semicontinuous sensing, seizing, and transforming is essential if the firm is to sustain itself as customers, competitors, and technologies change (Teece, 2007).

The continuous renewal dimension of dynamic capabilities is not unlike aspects of Nonaka’s (1994) knowledge spiral. In Nonaka’s theory of knowledge creation, individual tacit knowledge is shared (socialization), then made explicit (externalization) and synthesized with newly explicit knowledge from others (combination). This in turn gives rise to new knowledge as part of a learning process that creates new tacit knowledge (internalization), and so on in a knowledge spiral.

Dynamic capabilities reside, in part, with individual managers and the top management team.

⁶ Nonaka and Takeuchi (2011) noted that, unless companies create social as well as economic value, they will not survive in the long run. They quote former Toyota president Eiji Toyoda: “Doing the right things, when required, is a calling from on high” (Nonaka & Takeuchi, 2011, p. 62).

This was recognized early on by Adner and Helfat (2003). At certain critical junctures, the ability of a CEO and the top management team to recognize a key development or trend, then delineate a response and guide/lead the firm in its path forward, might be the most prominent feature of the firm's dynamic capabilities. But the organization's values, culture, and collective ability to quickly implement a new business model or other changes are also integral to the strength or weakness of the firm's dynamic capabilities. As Bartlett and Ghoshal (2002, p. 14) put it: "As major global competitors achieve parity in the scale of their operations and their international market positions, the ability to link and leverage knowledge is increasingly the factor that differentiates the winners from the losers and survivors." The reason is that knowledge and capabilities are not only scarce; they are often difficult to imitate. Sometimes they can be bought; generally, they have to be built. Their astute orchestration requires entrepreneurial capabilities that many management teams simply don't have.

Managerial decisions help create and shape capabilities and determine how they are deployed (Dosi, Faillo, & Marengo, 2008). This "asset orchestration" function (Teece, 2007), the core of dynamic capabilities, is undergirded by three sets of organizational processes (Teece et al., 1997): (1) coordination/integration, (2) learning, and (3) reconfiguration.⁷ Coordination and integration routines involve combining various resources in an entrepreneurial fashion, such as for the development of new products. Learning is an outcome of practice and experimentation and allows tasks to be performed more effectively. Reconfiguration, or transformation, entails recombining and modifying existing resources.

Although processes are an element of dynamic capabilities, good managers are able to think creatively and act entrepreneurially without the benefit (or constraint) of routines. To the extent that top management processes are routinized, they are likely to focus on "signature processes" (Gratton & Ghoshal, 2005). Signature processes arise from a company's heritage, including its prior management actions, certain irreversible investments, and context-specific learning. Because of their deep, enterprise-specific roots, they are not so easily im-

itated by other firms that did not and cannot share this history and that have a different corporate culture. Moreover, the replicability of a signature process or business model is often confounded, particularly externally, by what Lippman and Rumelt (1982) called "uncertain imitability." This, along with a high tacit component to the underlying knowledge, may keep a signature process effectively proprietary for considerable periods. As a result, signature processes themselves could satisfy the key criteria defined by Barney (1991) for resources that can support durable competitive advantage: valuable, rare, imperfectly imitable, and non-substitutable (VRIN). Hence, signature processes (and signature business models) can be an important source of heterogeneity, at least for quite a while (Jacobides & Winter, 2012).

A corollary of the fact that signature processes and certain other VRIN resources are products of the firm's heritage and past managerial decisions is that dynamic capabilities tend to get built, are difficult to imitate, and cannot generally be bought. Tim Cook, a longtime executive at Apple and its current CEO, said in February 2013, with reference to the company's integration of hardware, software, and services: "Apple has the ability to innovate in all three of these spheres and create magic. . . . This isn't something you can just write a check for. This is something you build over decades" (AFP, 2013). Cook's statement also hints at the interdependence between innovation in all three spheres.

Over longer periods, even signature processes become imitable by others. This transformation occurred with Toyota's lean production model, which is a tightly integrated set of processes that encompasses the entire value chain from product design to customer relations (Womack, Jones, & Roos, 1990). The "Toyota Production System" provided the automaker a source of competitive advantage for decades despite numerous and sustained attempts at imitation. However, it eventually diffused to other firms and industries. The same may occur in the development of new drugs, where an important goal is achieving regulatory approval. At present, many major pharmaceutical companies have well-developed, idiosyncratic systems for managing the approval process. In time, however, such systems could become standardized and available from a business service provider. When this occurs, what was a higher-order capability in the pharmaceutical industry degrades into a lower-order (ordinary) capability. This is the reason that strong dynamic capabilities require that the organ-

⁷ In terms of the three primary clusters of dynamic capabilities, asset orchestration is most relevant as an underpinning for seizing and transforming.

ization have the capability to modify (or reinvent) even signature processes, and to do so continuously, or at least semicontinuously.

The Basic Logic of the Capabilities Framework

Dynamic capabilities do not operate alone. They must be coupled with effective strategizing to bring about competitive advantage. The basic logic of the capabilities framework is shown in Figure 1.

In the diagram, organizational capabilities drive enterprise performance. They shape and are undergirded by VRIN resources. Strong dynamic capabilities must also be integrated with good strategy to effectuate strong performance. In short, the joint presence of strong dynamic capabilities, VRIN resources, and good strategy is necessary and sufficient for long-run enterprise financial success.

Strong ordinary capabilities (operations, administration, governance) must be accessed by the enterprise, but they need not necessarily be owned. Managing a plethora of ordinary capabilities can undermine dynamic capabilities. In other words, ordinary capabilities are not sufficient for long-

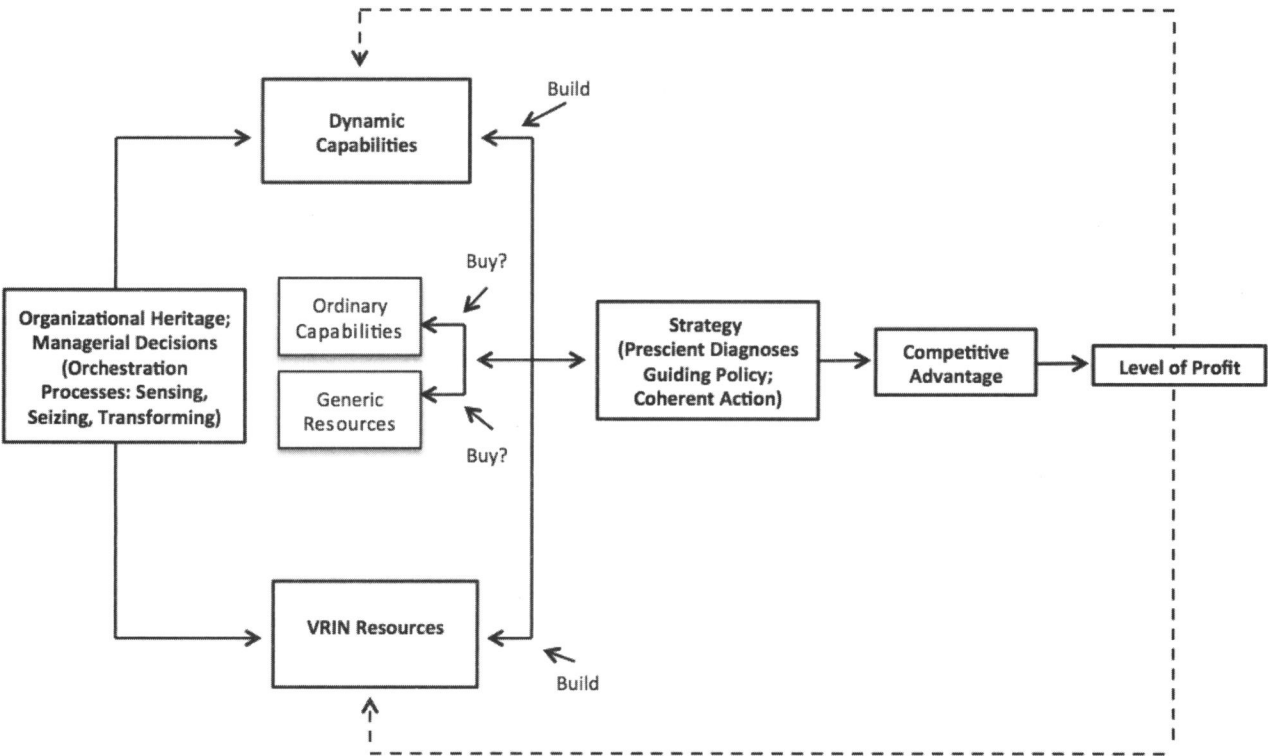
term financial success; they may not even be necessary.

Firms with weaker capabilities require different strategies than firms with stronger capabilities. For example, in a company that has a culture resistant to change, the new capabilities required to implement a strategy may need to be acquired rather than developed in-house, and the resulting acquisitions managed separately in a way that minimizes conflicts with incumbent units.

Ultimately, good performance requires strong dynamic capabilities to sense, seize, and transform in conjunction with a good strategy. It is important to emphasize the corollary: The effectiveness of dynamic capabilities can be compromised by poor strategy. The greater the diversity and rate of change in business environments, and the greater the importance of intangible (including relationship) assets, the more critical good strategy and strong dynamic capabilities become for the firm's growth and financial performance.

In the dynamic capabilities framework, the empirical phenomena being described are recognized as quite fluid, even if the underlying concepts

FIGURE 1
The Logical Structure of the Dynamic Capabilities Framework



Source: Based on Teece (2014).

are not. There are few constants; management theory is not physics. Thus, the aims of a positivist methodological program (ranging from naive verification through to Popperian falsification) are difficult, if not impossible, to fulfill in management studies. Deductive logic is a more useful approach. “Appreciative” (qualitative, observation-based) theorizing is perhaps the best way of characterizing the methodology of choice (Nelson & Winter, 1982).

For the theoretician, the dynamic capabilities framework requires deductive and inductive reasoning based on introspection and observation to develop the framework and, possibly, a theory of value creation and appropriation by firms. Clearly, there is much accepted theory (e.g., evolution and natural selection) that cannot be empirically verified or falsified (Rosen, 1997). Indeed, as Hayek (1989) reminded us, there are many concepts of paramount economic importance that cannot be measured empirically or “proved” scientifically.⁸

Some scholars (e.g., Arend & Bromiley, 2009) seem to believe that the framework is tautological. This is not the case, as the discussion of Figure 1 makes apparent. Long-run growth and profitability require the presence of strong dynamic capabilities, but the reverse is not true. Strong dynamic capabilities can become worthless if they are tied to a poor or badly misjudged strategy, and vice versa.

Empirical studies are appearing that provide support for the framework. These often take an in-depth case study approach (e.g., Danneels, 2011; Tripsas & Gavetti, 2000), but some studies are finding traction with large datasets (e.g., Adner & Helfat, 2003). In a recent example, Bardolet, Lovallo, and Teece (2014) framed cash reallocation across businesses as a dynamic capability and demonstrated its impact on financial performance.

The dynamic capabilities framework is subject to empirical examination at many levels of analysis, as other articles in this symposium demonstrate. For example, Schilke (2014) provides useful evi-

dence on how dynamic capabilities are built and maintained by showing that alliance learning, a second-order capability, shapes both the alliance management capability and alliance performance. Also in this issue, Kleinbaum and Stuart (2014) dig down into the interpersonal networks within an organization that can determine the network’s responsiveness to structural change.

Inertia, Transformation, and Learning

Dynamic capabilities include standard entrepreneurial activities such as identifying undersatisfied demands and mobilizing resources in such a way as to profit from satisfying them. But the critical value of dynamic capabilities lies in their potential for helping the organization do this repeatedly, thereby helping to create a durable competitive advantage.

The “transforming” aspect of dynamic capabilities is needed most obviously when radical new opportunities are to be addressed. But transformative capabilities must also be exercised periodically to soften the rigidities that develop over time from asset accumulation, standard operating procedures, and insider misappropriation of rent streams. A firm’s assets must be kept in strategic alignment vis-à-vis its ecosystem. Complementarities need to be constantly managed (and reconfigured as necessary) to achieve evolutionary fitness, such as by limiting loss of value when market circumstances shift to favor external complements. The strength of a firm’s dynamic capabilities determines the speed and degree to which the firm’s resources can be aligned consistent with the firm’s strategy and with changes in the opportunities and requirements afforded by the business environment. Transformation and asset reconfiguration depend critically on the top management team’s ability to lead—that is, to persuade others to undertake different courses of action and adopt different states of mind (Helfat & Peteraf, 2014).

The idea that organizations are potentially able to adapt has deep roots in the literature. March and Simon (1958) recognized that, in principle, organizations can design processes to bring new programs into existence and later modify them. Indeed, the Carnegie School always recognized adaptation, which implies that change occurs, albeit slowly (Cyert & March, 1963).

The potential transformation envisioned when an enterprise has strong dynamic capabilities goes beyond narrow notions of “strategic fit” seen as

⁸ Some scholars might argue that, without formalism, there are no theories. However, this viewpoint cannot withstand logical or historical scrutiny. There are many fields (e.g., geology) that are not formal yet gained the status of a theory-based science when interesting causal structures, which were not necessarily formal, were advanced (e.g., the theory of plate tectonics). Both formal and ordinary language can be used rigorously or nonrigorously. Formality ought not be the standard; rigor can be.

optimal in the “adaptation” school of organizational change research. That school sees the environment as exogenous. Teece (2007) and Sirmon, Hitt, and Ireland (2007) emphasized a “fit” that is far broader and embraces (1) the firm’s internal processes, (2) partners, (3) customers, and (4) the business environment. Achieving tight “fit” with all four is likely to require strong dynamic capabilities.

The managerial message from the dynamic capabilities approach is virtually orthogonal to the rather bleak managerial message from the “organizational ecology” school of research (e.g., Hannan & Freeman, 1977). Organizational ecologists hold that, when the business environment shifts, incumbent firms face overwhelming inertia due to micro-political social practices, “bargains,” and/or relational contracts within the company and are, as a result, displaced by organizations better suited to the changed context. Although there is evidence of organizational inertia, the dynamic capabilities framework holds that management can overcome (i.e., “manage”) evolutionary forces to some degree (O’Reilly & Tushman, 2008). The changes that have occurred in the course of the histories of numerous major corporations, such as GE, IBM, Nokia, Netflix, and Apple, suggest that this is true in practice. However, further research is needed to reveal more about how and when this can be accomplished, and the manner and cost of doing so.

Incremental change by firms is, in fact, relatively common.⁹ More radical change is possible, but it requires organizational mutation. Such mutation must be engineered and driven by entrepreneurial managers/leaders.

While path dependence poses a constraint on the future actions of all enterprises, for some firms the legacy of the past can also provide the foundation and fulcrum of future growth. In the modern parlance of Silicon Valley, firms must (and some do) “pivot” when inflection points occur in the ecosystem or when they discover that their strategy and/or business model isn’t working. It is, of course, easier for startups than for established enterprises, and it may be easier for Silicon Valley-type firms that have shallow hierarchies and strong

leaders (Teece, 1992). Indeed, the so-called “lean startup” methodology is all about effectuating rapid and continuous transformation until winning combinations are discovered and enacted (Ries, 2011). Anti-cannibalization instincts and entrenched interests often stand in the way for incumbent firms and undermine their dynamic capabilities. In other words, firms must address pressures and tensions caused by processes internal to the firm as well as those in the external business environment (Greiner, 1998).

Organizational sclerosis can manifest itself as a failure to change investment patterns or a failure to change routines (Gilbert, 2005). At the core of most rigidities are business models—and managers unable to recognize the need to modify or abandon them.

In the dynamic capabilities framework, routines and business models are not viewed as completely wooden. It is true that routines are bound by rules, customs, and heritage, yet they are capable of being changed in most instances—especially with strong leadership. Some companies, such as Netflix under Reed Hastings, have demonstrated a remarkable ability to adopt new business models midstream, but not without considerable investor angst and customer confusion.

If an enterprise is to have strong dynamic capabilities, it must not only ensure that strategy and organization are aligned with anticipated changes in markets, technologies, and the business environment, it must also be able to effectuate change in ordinary capabilities—and their underlying routines. In some cases, it may suffice simply to access different suppliers. At other times, more radical measures may be required.

As Jack Welch, the storied former CEO of GE, is widely quoted to have said, “If the rate of change inside an organization is less than the rate outside, the end is in sight.” However, having the imperative to change is not enough. Framing a clear vision and finding the migration path from the old to the new, including figuring out which “rules” to break in order to get there, are likely to be required.¹⁰ If done well, the result will be new, high-performance signature processes.

⁹ Augier and Teece (2008) pointed out that the dynamic capabilities framework involves “evolution with design.” However, that article conflates dynamic capabilities with “strategic processes.” In the present article, I am drawing a distinction between the capabilities and strategy concepts.

¹⁰ As Peter Drucker is widely quoted to have said: “If you want something new, you have to stop doing something old.” But this leaves managers to fill in the messy details of what to modify and what to leave as is.

Some organizational change is reactive. As Winter (2003, p. 993) explained, firms get into “firefighting mode,” which he characterized as a “high-paced, contingent, opportunistic and perhaps creative search for satisfactory alternative behaviors.” Winter called this ad hoc problem solving—in contrast to routine-directed problem solving.¹¹ Yet he correctly recognized that it is possible that on close examination even “firefighting” approaches to problem solving may have micro-routines embedded within.

Productive transformation, the kind that leads to enterprise growth, ultimately requires routines and entrepreneurial actions for the introduction of new activities alongside existing business lines. In too many organizations, there may be a readily available set of routines for downsizing the enterprise, but there is generally no routine in place for growing it.

In the course of exploring future possibilities while exploiting present opportunities, many potentially incompatible activities will need to take place simultaneously: developing new products and services while improving existing ones, servicing existing customers while acquiring new ones, hiring top talent while retaining current staff, introducing new processes while improving operations, and so on. This requires a particular subset of dynamic capabilities that O'Reilly and Tushman (2004) called “ambidexterity.”

Common to the pursuit of change, the fine-tuning of existing (ordinary) capabilities, and the synchronization of business processes and models with the business environment is learning, a critical dimension of dynamic capabilities. The enterprise must learn (1) what customers want, (2) what new technologies might allow, (3) what aspects of the business model are working, and (4) whether the current strategy is effective and the company is on the path toward building a great business. It should be

noted that there is nothing automatic about technological and organizational learning.

Learning and innovation benefit from light-touch management—especially where expert talent is concerned (Teece, 2011b). Horizontal interaction among talented people with diverse knowledge bases inside and outside the enterprise is generally required to solve complex problems, giving support to the “open innovation” paradigm (Chesbrough, 2003). Dynamic capabilities demand both an external (outside the organization) and internal orientation by management. The learning and innovation that undergird transformation (and dynamic capabilities more generally) and contribute to durable competitive advantage often need to be global in scope. Multinational enterprises competing in diverse contexts have the possibility to learn across different geographies (Teece, 2014).

The dynamic capabilities framework could also be applied to help illuminate national economic development, such as the successes of the Asian “tigers” (and the lackluster outcomes in many other countries). Whereas traditional economic development theorists stress resource accumulation (propelled by high rates of investment), the dynamic capabilities framework stresses the importance of enterprise-level entrepreneurship, innovation, learning, and good strategy. This resonates with emerging theories of development (Lall & Teubal, 1998). When Nelson and Pack (1999, p. 434) noted that “if . . . one marshals [inputs] but does not innovate and learn, development does not follow,” they implicitly endorsed the importance of dynamic capabilities for national economic development.¹²

THE BIFURCATION OF THE LITERATURE: INTEGRATING AND RELATING DISPARATE ELEMENTS

I have so far elaborated and expanded the logic of the Teece and colleagues (1997) and Teece (2007) framework. In this section, I will more directly address the bibliometrically defined bifurcation in the dynamic capabilities literature identified by Pe-

¹¹ Processes can be both routinized and ad hoc. In an interview about product development at Apple (Burgelman, 2004), the company's late CEO, Steve Jobs, described it as a blend of routine and creative acts: “Apple is a very disciplined company, and we have great processes. But that's not what it's about. Process makes you more efficient. But innovation comes from people meeting up in the hallways or calling each other at 10:30 at night with a new idea, or because they realized something that shoots holes in how we've been thinking about a problem.”

¹² Nelson and Pack (1997) distinguished between accumulation and assimilation theories of development. The assimilation approach aligns with dynamic capabilities theories of the development and growth of the business enterprise. The accumulation approach is more akin to the resource-based view of the firm.

teraf and colleagues (2013), which they traced to Eisenhardt and Martin (2000).

Teece and colleagues (1997) claimed that dynamic capabilities are important for sustainable firm-level competitive advantage, especially in high-velocity markets. Conversely, Eisenhardt and Martin (2000) claimed that dynamic capabilities are inherently unsuited to creating sustainable advantage and that they are likely to break down in high-velocity markets.

Writing on these contrasting views, Peteraf and colleagues (2013) argued that Eisenhardt and Martin, in effect, endeavored to reconceptualize dynamic capabilities, “challenging the purpose and mechanisms of the [Teece et al.] framework and delimiting its boundary conditions” (p. 1391). I agree with their assessment. While there can be benefits in reconceptualization, the reconceptualization here compromised essential elements in the initial (Teece et al.) framework.

To clarify key distinctions between the two understandings of dynamic capabilities, I will start by sketching the Eisenhardt and Martin logic. First, Eisenhardt and Martin (2000, p. 1107) defined dynamic capabilities as “the organizational and strategic routines by which firms achieve new resource configurations.” Next, they claimed, somewhat surprisingly, that all dynamic capabilities can be captured as “best practice.” Accordingly, dynamic capabilities are seen as imitable by rivals, and so cannot be a source of competitive advantage. Finally, in high-velocity business environments (where Teece and colleagues, 1997, claimed that the dynamic capabilities approach is particularly relevant), Eisenhardt and Martin (2000, p. 1113) claimed that the processes of dynamic capabilities are limited to simple rules that are less stable than fixed routines and are likely to “collapse.”

Peteraf and colleagues (2013) tried to reconcile this version of dynamic capabilities with that of Teece et al. within the limitations of the assumptions stated in each article. It should also be acknowledged that the Teece and colleagues (1997) framework has continued to evolve. In fact, the Teece et al. (1997, p. 515) paper clearly indicated that it was just the first step in fleshing out a framework, noting at one point that a goal of its exposition was “to facilitate theory development and intellectual dialogue.” There are several articles that elaborate the Teece et al. view of dynamic capabilities in ways that endeavor to help clarify the issues at hand (e.g., Teece, 2007, 2012b, 2014).

The divide between the Eisenhardt cluster and the Teece cluster of dynamic capabilities research can be bridged, according to Peteraf and colleagues (2013), by various approaches. Perhaps the simplest is to recognize that Eisenhardt and Martin are, in the main, focused on a different class of capabilities, those I described earlier as ordinary capabilities, which can be benchmarked for best practice and are vulnerable to imitation. This is a major point of difference. As noted in Teece (2007, p. 1321), “a well understood and replicable best practice is not likely to constitute a dynamic capability.”

In the remainder of this section, I will begin by exploring the role of managers, another key distinction between the Eisenhardt and the Teece conceptualizations of dynamic capabilities. Managerial action is a critical complement to organizational routines within the Teece dynamic capabilities framework. I will also discuss the role of VRIN resources and the relationship between dynamic capabilities and strategy, before returning briefly to the central importance of the distinction between ordinary and dynamic capabilities.

The Role of Managers

First, I reject the notion that dynamic capabilities reside *only* in high-level routines. Teece et al. (1997, p. 516) defined dynamic capabilities as “the firm’s ability to integrate, build, and reconfigure internal and external competences to address rapidly changing environments.” I have endeavored to make clear that, in my view, dynamic capabilities involve a combination of organizational routines and entrepreneurial leadership/management (Augier & Teece, 2009; Teece, 2007, 2012b). Creative managerial and entrepreneurial acts (e.g., creating new markets) are, by their nature, often non-routine. Indeed, “many strategic actions and transformations require actions that one may never replicate” (Teece, 2012b, p. 1397). The differentiated experience of Yum! Brands’ KFC units in Japan and China would appear to bear this out. KFC Japan became a success in the 1970s under the leadership of Loy Weston and Shin Ohkawara, who developed the local branch more as a fashion business than as fast food (Bartlett & Rangan, 1986). In China, however, KFC’s rapid growth during the 1990s is generally attributed to the aggressive store build-out and radical menu localization that took place under the leadership of Sam Su (*China Economic Review*, 2011).

Leadership matters with respect to the organization's ability to effectuate discovery, co-creation, and change. To modify and extend the metaphor that Di Stefano et al. introduce in their article in this issue, dynamic capabilities are not just the gears and drivetrain of the organizational machine; they include the entrepreneurial consciousness, imagination, and human action that guide it. The human action involves sensing, creating and co-creating, seizing, and transforming.

A theory of competitive advantage without the possibility of non-routine action by the top management team is awkward, and the dynamic capabilities framework was never meant to preclude non-routine action. Eisenhardt and Martin's (2000) article demonstrates an acute awareness of the role of managers, such as their "emotional inability to cope with uncertainty . . . in high-velocity markets" (2000, p. 1112). Yet their argument about the inherent weakness of dynamic capabilities theory in high-velocity environments uses only the language of processes and rules.

While no firm will succeed forever in a particular market, strong dynamic capabilities allow a firm to ride successive waves of change across lines of business by renewing and leveraging the (fungible) services of their valuable and difficult-to-replicate resources. Texas Instruments (TI), for example, went from being a diversified electronics producer in the 1980s to having a narrower focus on memory chips in the 1990s, switched in the 2000s to digital signal processors (DSPs) for cell phones and other devices, and now has refocused yet again on analog chips. These transitions, each of which drew on existing resources within the company, occurred under a series of CEOs who were all career-long TI employees. Over the turbulent period 1992–2012, the company averaged a net income margin of nearly 12% of sales despite the downdraft of the dot-com bust and other challenges. The strong, shared values of TI's leaders and employees, the firm's technological capabilities, and its ability to use its technology and intellectual property astutely underpin the dynamic capabilities that have helped the firm to thrive in the very turbulent global electronics industry.

In high-velocity environments, the business enterprise may well be particularly reliant on the sensing and seizing instincts and actions of the CEO and the top management team. To the extent that this is so, the capabilities will, of course suffer from a degree of instability because their longevity depends logically on the tenure of entrepreneurs/

managers/leaders. Over time, however, rules and processes to simulate and perpetuate the intuition behind some elements of a good leader can be developed. At Apple, for example, Steve Jobs recognized the risk of having the firm be a one-man show and set up a corporate university where executives teach professionally prepared cases about how key past decisions, such as the creation of the Apple Store, were reached (Lashinsky, 2011). In this way, the values and the approach of the founders are disseminated among Apple's current and future leaders. Similarly, Andy Grove helped keep Intel agile with his constant reminder that "only the paranoid survive" (Grove, 1996).

Eisenhardt and Martin (2000) held that dynamic capabilities would break down in high-velocity environments because of the instability of the simple rules (basically, semi-improvised managerial actions). It is clearly possible, however, for some companies to sail through a variety of high-turbulence markets relatively unscathed, as TI did first with memory chips, then with software-intensive chips for cell phones.

Of course, the notion of "simple rules" does not necessarily preclude a decision-making role for managers. As Di Stefano et al. point out in this issue, simple rules "imply that the *agency* rests in individual managers."

Once we readmit one-time entrepreneurial/managerial actions (with or without rules) as a component of dynamic capabilities, the logic by which Eisenhardt and Martin reasoned that dynamic capabilities are unstable in high-velocity environments is undermined. Yes, high-velocity environments are challenging and require, as Eisenhardt and Martin (2000, p. 1113) so aptly noted, "extensive and frequent use of prototyping, real-time information, experimentation, and multiple alternatives." But that is precisely the type of managerial responsiveness the dynamic capabilities framework envisages. Strong dynamic capabilities in high-velocity environments are not stable so much as protean, potentially allowing any organization to retain the "pivot" potential of a "lean startup" (Ries, 2011). Admittedly, small startups can "pivot" more readily than can large established enterprises.¹³

¹³ However, Lou Gerstner demonstrated with IBM that it is possible to make elephants (i.e., large firms) dance (Gerstner, 2002).

Resources and the VRIN Criteria

A dynamic capability involves building and orchestrating resources to perform a changing kaleidoscope of tasks and activities. An understanding of the role of resources is therefore important to a proper understanding of the dynamic capabilities framework.

Resources are potentially productive tangible and intangible assets and people that are semipermanently attached to a firm. In the resources framework developed by Penrose (1959) and then re-framed by Richardson (1972) in the related language of capabilities,¹⁴ the firm's managers can combine services yielded by resources in several ways.¹⁵ At any point, management must choose which market opportunities the firm will pursue with the resources on hand.

In earlier work, Teece (1980, 1982) explored Penrose's ideas of resource fungibility by assessing how the nature of a resource, in particular its "tradability" (or lack thereof), affected the nature of enterprise growth through diversification. Opportunities to leverage non-tradable assets (generally know-how) across different lines of business propelled the creation of diversified multiproduct firms. These early treatments focused on how transaction costs might affect diversification strategy. They did not, however, give systematic attention to how non-tradable assets and resources get created, protected, and deployed. They also neglected the role of (entrepreneurial) management. These deficiencies have been remedied by the dynamic capabilities framework.

While Penrose recognized the importance of productive opportunities and entrepreneurship, she did not fully address the importance of resources for building competitive advantage. Wernerfelt (1984) and Barney (1986, 1991) partly filled this gap by focusing on the possession of certain resources as the main mechanism for anchoring the generation of economic rent. The resources of strategic interest were defined by Bar-

ney (1991) as those satisfying the VRIN criteria, defined above.

Intangible resources in general, and intellectual capital in particular, are the class of assets that, in my view (Teece, 2000), most frequently meet the VRIN criteria. These were the focus of some of my early research. Intangible resources are difficult to trade in most cases because their property rights are likely to have fuzzy boundaries and their value is context-dependent (Teece, 1981). The primary exceptions are codified knowledge for which transferable rights have been conferred by law, such as patents, trademarks, and copyrights, and even these face high transaction costs, such as those for monitoring infringement and getting recalcitrant infringers to take a license. There is unlikely to be a well-developed market for most intangibles, and they are also generally difficult to transfer from one firm to another because the transfer of many intangibles, such as trade secrets, often entails the transfer of people.

The resources approach implies that competitive advantage flows from heavy endowment or investment in the right (i.e., VRIN) resources. But even high-potential resources must be astutely managed (orchestrated) to boost enterprise performance. Managerial coordination of resources is not featured in the resource-based approach, but is critical in dynamic capabilities.

The asset orchestration function is critical to organizational performance in the military as well as in business. General Stanley McChrystal has made it clear that, during the period of the United States Army's involvement in Iraq, resources alone were not enough: "We had a culture in our forces, of excellence. It was how good can I be at my task? . . . But that's not as important as how well those pieces mesh together. . . . The real art is . . . cooperating with civilian agencies, it's cooperating with conventional forces, it's tying the pieces together. That's the art of war, and that's the hard part" (Rose, 2013, p. 6).

In short, VRIN resources, in and of themselves, are inherently valuable by definition, but they do not generate long-term enterprise value (or military prowess) on their own.¹⁶ For long-term growth and survival of the enterprise, they must be

¹⁴ "Mrs. Penrose has provided us with excellent accounts of how companies grow in directions set by their capabilities and how these capabilities themselves slowly expand and alter" (Richardson, 1972, p. 888).

¹⁵ Penrose's framework includes "the range of ideas of entrepreneurs among the services rendered" (Penrose, 1959, p. 86). In this regard, Penrose was perhaps describing a dynamic capability.

¹⁶ Firms with strong dynamic capabilities will likely possess VRIN resources. Some scholars may prefer to think of dynamic capabilities themselves as being a long-term VRIN resource. I find this confusing as it distracts from the asset (resource) orchestration function that is core to dynamic capabilities.

cleverly managed, or orchestrated, by a dynamically capable management team pursuing a good strategy. This also means that the resource-based view of the firm needs dynamic capabilities to explain how assets get deployed and how rent streams get extended and renewed.

The resources framework also puts little weight on entrepreneurship, innovation, or learning. Indeed, as Barney and Clark (2007) noted, “resource-based theory takes the existence of heterogeneous firm resources and capabilities as given and examines the impact of the resources for the ability of firms to gain and sustain competitive advantage” (2007, p. 257).

The dynamic capabilities framework endeavors to go further by exploring various “microfoundations.” The framework addresses (1) innovation and co-creation, (2) where resources come from (i.e., how they can be built), and (3) how resources should be deployed. It approaches these issues by appealing to strategy (for determining the resource allocation targets) and to organization structure and processes (to facilitate resource/asset redeployment).

Dynamic Capabilities and Strategy

As discussed earlier, dynamic capabilities must be used in aid of a good strategy to be effective. Strategy, capabilities, and the business environment co-evolve. A strategy that is consistent, coherent, and accommodating of innovation is needed to help achieve competitive advantage. A firm with strong dynamic capabilities is able to flesh out the details around strategic intent and to implement strategic actions quickly and effectively.

A strategy can be defined as “a coherent set of analyses, concepts, policies, arguments, and actions that respond to a high-stakes challenge” (Rumelt, 2011, p. 6). For Rumelt, a good strategy has (1) a diagnosis, (2) a guiding policy, and (3) coherent action. It is clear that this taxonomy interacts with the three clusters of dynamic ca-

pabilities: sensing, seizing, and transforming. Sensing contains a strong element of diagnosis, which is important to strategy. Seizing needs to be connected to both a guiding policy and coherent action. Transforming that protects and enhances value requires a guiding policy and coherent action. The nature of the managerial tasks for various elements of strategy and dynamic capabilities is outlined in Table 2.

Earlier, I described how TI demonstrated strong dynamic capabilities over decades with a series of major product market transitions. But strategy was also implicated in each of these changes. Divesting a major line of business such as memory chips and focusing on DSPs involves calibrating opportunities and realigning resources, but it also involves strategic analysis such as how best to capture value from the firm’s intellectual property portfolio (Grindley & Teece, 1997).

Strategy and dynamic capabilities both occur at the line-of-business as well as the company-wide level. This is consistent with other views of modular organizations (e.g., Helfat & Eisenhardt, 2004), where company-wide strategy is different and separable from the day-to-day positioning strategy at the division level. Strategy, when developed successfully, provides the specifics of how the firm will deploy its scarce assets to support market needs and outmaneuver rivals. Strong dynamic capabilities provide the flexibility to make the necessary adjustments. As Lou Gerstner said at the start of his tenure as CEO of IBM, “You have to be fast on your feet and adaptive or else a strategy is useless” (quoted in Sellers, 1993). He might as well have said that strategy is useless without strong dynamic capabilities.

Put differently, VRIN resource accumulation and the managerial orchestration discussed above, which is central to enhancing processes and exploiting positions, must be guided and informed by strategy. Frank Hoffman’s (2004) analysis of the British Navy’s inability in 1916 to win the Battle of Jutland seems relevant here. Despite the British

TABLE 2
The Interrelation of Dynamic Capabilities and Strategy

Strategy kernel	Diagnosis	Guiding policy	Coherent action
Related dynamic capabilities schema	Sensing	Seizing/transformation	Seizing/transformation
Nature of managerial orchestration	Entrepreneurial	Administrative	Leadership

Source: Teece (2014).

Navy's numerical advantage, the battle was a stalemate.¹⁷ Both sides claimed victory, but neither achieved it. British Vice Admiral Sir David Beatty ruefully proclaimed at the time that "there seems to be something wrong with our bloody ships today" (which can be interpreted as surprise that superior British naval resources did not prevail) (Butler, 2006, p. 151). Hoffman (2004, p. 70), reviewing the situation nearly a century later, concluded that "[t]he real deficiency, however, was the loss of [Vice Admiral Horatio Lord] Nelson's touch. It was not the bloody ships that were principally at fault. It was the inadequate doctrine of command and control." Put differently, the British failure to leverage their superior resources into a victory reflected a failure in (military) strategy and dynamic capabilities.

Ordinary Capabilities and Best Practices

As noted above, Eisenhardt and Martin's (2000) article misinterpreted (or reframed) the dynamic capabilities framework by claiming that all capabilities, including dynamic capabilities, can ultimately be characterized by best practice and hence imitated.¹⁸ In essence, as now explained in more detail, Eisenhardt and Martin conflated two concepts that benefit from being analytically separated, namely ordinary and dynamic capabilities. Ordinary and dynamic capabilities are, as discussed earlier, quite distinct, both analytically and in practice (see Table 1).

To recap, ordinary capabilities involve the performance of those administrative, operational, or governance-related functions that are (technically) necessary to complete currently planned tasks. Ordinary capabilities permit a firm to get things done and "make a living" (Winter, 2003, p. 991), albeit possibly a mediocre one.¹⁹ In mathematical terms,

ordinary capabilities can take on only zero or positive values; a firm can either perform an activity to some level of proficiency or it can't.

As explained earlier, ordinary capabilities, including best practices, are not sufficient to drive sustainable competitive advantage. "Leveling up" (capability equalization among competitors) is often possible with respect to best practices (ordinary capabilities) because they can be learned. Even standard economies of scale can often be achieved through heavy investment. Strong global competition, technology transfer activities, and operations improvement programs led by management consultants all help with leveling up.

While ordinary capabilities and associated best practices are far from fully diffused globally (Bloom et al., 2012), it is easier than ever before to imitate best practices.²⁰ A good deal of know-how that was tacit and proprietary just two or three decades ago is now explicit and even in the public domain—available from textbooks, consultants, schools of engineering, and schools of business. Explicit (codified) knowledge travels easily, and the Internet, by allowing low-cost connectivity, has served to accelerate long-established trends. Because of their imitability, ordinary capabilities are insufficient for long-term survival and growth, except in very stable and protected business environments (Drnevich & Kriauciunas, 2011).

Ordinary Capabilities, Dynamic Capabilities, and Inimitability

Dynamic capabilities, by contrast, are extremely difficult to equalize across companies. They do not

¹⁷ The British (aided by the Australians and Canadians) had 151 combat ships, including 28 battleships. The Germans had 99 combat ships, including 16 battleships. Some commentators are convinced that the British missed a tremendous opportunity to annihilate the German fleet and win what would have been another Trafalgar.

¹⁸ Peteraf et al. noted quite correctly that just because rivals can imitate, to some extent, various capabilities does not mean they will be "equally adept" (2013, p. 1403) or that competitors level up instantaneously.

¹⁹ Strong ordinary capabilities can lead, in weak selection (competitive) environments, to steady prof-

its, at least until competition grows stronger. However, even though the enterprise itself may generate only modest profits, significant employment can be generated. Nation-states may get strong benefits from having business enterprises with weak dynamic capabilities dominate the domestic economy even when shareholders don't benefit much or at all. The inverse may also be true: Enterprises with strong dynamic capabilities will generate good shareholder returns but not necessarily high employment.

²⁰ Google identified eight management "behaviors" (practices) that, based on employee interviews, lead to more effective management. However, as Garvin (2013, p. 82) noted, "the eight behaviors . . . might not give Google a lasting edge. Companies with similar competitive profiles . . . can mimic Google's approach, since the eight behaviors aren't proprietary."

typically level up. One reason is that they may be grounded in signature processes, rooted in each company's unique history and personnel. Uncertainty about the true causal linkages makes them hard to imitate, even with equifinal, much less duplicate, processes. Key dynamic capabilities, such as prescient identification of new market opportunities, involve evaluation, assessment, and judgment that are hard to codify and, accordingly, difficult to imitate. Dynamic capabilities are strong when leadership is wise. Indeed, dynamic capabilities might actually be what undergirds "phronetic" leadership (Nonaka & Toyama, 2007).²¹

Dynamic capabilities can be weak as well as strong, a distinction that appeared in Helfat and colleagues (2007) and Teece (2007). Weak dynamic capabilities can actually undermine a firm's prospects. Indeed, inertia may be preferable to taking a deeply misguided action. In mathematical terms, dynamic capabilities can take on not only positive but also negative values.

Dynamic capabilities are partly resident in the leadership team itself, rendering such capabilities even more idiosyncratic. Ordinary capabilities require good administration and management. While ordinary capabilities matter for enterprise performance, they lack the long-term impact of the (leadership-dependent) change management that is a vital part of strong dynamic capabilities.

As noted, most of the capabilities identified by Eisenhardt and Martin (2000) as dynamic capabilities, such as product development and alliance formation, can more properly be thought of as ordinary capabilities. Perhaps this accounts for their claim that these capabilities don't satisfy the VRIN criteria and cannot sustain competitive advantage. As noted in Table 1, ordinary capabilities support *technical* fitness—that is, "doing things right." Only dynamic capabilities support *evolutionary* fitness, which has more to do with innovation and "doing the right things"—and with continuing to do so as the business environment shifts (Teece, 2009, p. 7).

Bob Lutz (2011) of General Motors put this aspect of dynamic capabilities rather succinctly for the auto industry:

Where the real work of making a car company successful suddenly turns complex, and where the winners are separated from the losers, is in the long-

cycle product development process, where short-term day-to-day metrics and the tabulation of results are meaningless.

Put differently, a firm's ordinary capabilities enable the production and sale of a defined (but static) set of products and services. However, ordinary capabilities do not help determine whether the current production schedule is the right (or even a profitable) path to follow in the future. When the firm's output is tuned to what the market desires, strong ordinary capabilities may be sufficient for a fleeting competitive advantage but are insufficient to undergird sustainable competitive advantage as the business environment changes.²² The "real work" of building an organizational advantage, to paraphrase Lutz, involves building and using dynamic capabilities.

ECONOMIC THEORY AND CAPABILITY THEORY

So far, I have addressed what I see as the main gaps between my view of dynamic capabilities and the related but distinct scholarship on dynamic capabilities that has evolved, according to Peteraf and colleagues (2013), around Eisenhardt and Martin (2000). In this section, I address the multidisciplinary nature of my framework. Since my view is most often associated with the economics discipline, I will primarily discuss how mainstream economic theory has assumed away any representation of capabilities even though the capabilities of firms undergird modern economic growth.

Peteraf and colleagues (2013, p. 1399) identified separate sets of disciplinary orientations for the "Eisenhardt group" of authors (organizational theory, science, or behavior; information systems) than for the "Teece group" (economics, strategy, technology). However, the involvement of scholars from different disciplines should not, in itself, be a

²¹ *Phronesis* is the Greek word for a practical type of intelligence or wisdom.

²² Multinational enterprises investing in less-developed economies constitute a special case. Firms domiciled in advanced economies "appear to adopt good management practices in almost every country in which they operate" (Bloom et al., 2012, p. 14). They may thus succeed for a while with strong ordinary capabilities, because ordinary capabilities developed at home may temporarily be distinctive abroad, where the competition or "selection" environment may be weak. The overseas success McDonald's has had with fast food is due in part to its ability to transfer its considerable ordinary capabilities.

problem. In fact, all these different strands have contributed to the development of the dynamic capabilities framework to some extent. As I have argued elsewhere, dynamic capabilities can serve as “a framework for integrating and enhancing the entire business school curriculum” (Teece, 2011a, p. 499).

A full understanding of how organizations and leaders exhibit strong or weak dynamic capabilities requires insights from many disciplines and sub-disciplines in the social sciences. A multidisciplinary approach is even more critical for an understanding of the deeper question of how business systems facilitate the integration of ideas and their conversion into actions that generate economic outcomes.

One should not be surprised if individual researchers tend to pursue research within the dynamic capabilities framework using the tools and perspectives that count in their respective disciplines. However, it is my hope that dynamic capabilities can serve as a meta-theory or framework that facilitates multidisciplinary collaboration in which each discipline will find a useful place.²³ To take one of the five domains of “bifurcation” identified by Di Stefano and colleagues (2014), a researcher who selects organizations as the primary focus and one who instead focuses on managers will be able to find common ground in the dynamic capabilities framework, which provides an understanding of the complementarity between these two levels of analysis within the larger perspective of the pursuit of firm-level competitive advantage.

Di Stefano and colleagues (2014) find that the dynamic capabilities literature most associated with Teece and colleagues (1997) relies more on economic logic than does the Eisenhardt and Martin-based cluster. While this is true, one should be aware that mainstream (neoclassical) economic theory is a long way from recognizing any concept like capabilities, despite its need to do so.²⁴ More-

over, even technical efficiency (the result of strong ordinary capabilities) is assumed, not established, in economic theory.

One of the few overlaps with mainstream economics is that the dynamic capabilities framework emphasizes the importance of inimitability. This is an economic concept of great importance to the capabilities framework. It also provides a framework for strategic focus; managers can unclutter their radar screens by prioritizing VRIN resources and, if necessary, outsourcing activities that depend on ordinary capabilities since these are of far less strategic importance due to their very outsourceability/acquirability.²⁵

In the remainder of this section, I examine the dynamic capabilities framework with respect to economic theory. The basic theme is that dynamic capabilities are not as well integrated with the economic theories of the firm and of markets as they should be—and are certainly less integrated with (mainstream) economic theory than some seem to think. However, there is compatibility with those elements of Austrian economics that stress the importance of innovation, uncertainty, and disequilibrium.

The significance of a capabilities framework is perhaps best understood when juxtaposed against the mainstream economic theory of resource allocation via markets. Some economists would have us believe that market exchange activity is the sole basis of resource allocation and wealth creation. In light of this, fundamental questions for organizational theory are these: Why are organizational capabilities needed in the economic system? And where do firm-level capabilities fit into the intellectual framework that economics employs?

Economics, we are told in almost every introductory textbook, is about the allocation of (scarce)

²³ For example, Di Stefano and colleagues (2014) describe how behavioral theorists favor one cluster of dynamic capabilities research over the other by studying organizational adaptation with limited concern for competitive position. However, that they do so while invoking dynamic capabilities does not weaken the dynamic capabilities framework. Instead, it strengthens behavioral theory by implicitly recognizing the broader requirements of competition and profitability within which adaptation occurs.

²⁴ Much of economic theory still adopts the homoge-

neity assumption with respect to firms. The primary exception is game theory, where differences (asymmetries) are recognized. However, most game theoretical models are decidedly non-robust, providing substance to Sutton's claim that, by proving everything, game theory proves nothing (Sutton, 1990).

²⁵ Strong dynamic capabilities are a prerequisite for effective outsourcing. Boeing set up the supply chain for its new 787 Dreamliner passenger jet using a broader array of global suppliers than it had previously. Unfortunately, it also cut back its monitoring capability. Problems with inadequate components led to a delay of more than three years (Kesmodel, 2011). This was a failure of asset orchestration.

resources (and goods and services) among unlimited wants.²⁶ The fundamental task is to get resources allocated efficiently so the available resources are used to satisfy the strongest demand (highest willingness to pay). The role the price system plays in doing this is generally referred to as allocative efficiency. Economists (e.g., Hayek, 1945) wax eloquent about how well the market does this kind of allocation, and appropriately so. But Hayek and other Austrian School economists such as von Mises and Kirzner maintain that people do not allocate means to ends; rather, they consistently seek to discover and create new ends and means. In this regard, Austrian economics is compatible with dynamic capabilities. Neoclassical economics is not.

It is important to recognize that, when scarce resources must be allocated among unlimited wants, more than the price system is needed to achieve this goal. Managers and management are needed too, particularly if innovation and change are implicated. The (neoclassical) economic model of market exchange takes for granted that somehow, somewhere, new goods and services are being designed, developed, and produced by some method that will be technically efficient, conditional on factor costs. Moreover, it is assumed that everyone knows about it. The transaction cost economics framework (Williamson, 1975, 1985) has similar (perhaps deliberate) blind spots. It holds “production” activity invariant even though production costs may depend endogenously on the governance modes, managerial actions, strategy, and structures chosen. Moreover, production technologies and governance modes are not, in the economic perspective, proprietary, but rather available to all firms. Omniscience, not ignorance, is the norm.

As both a theoretical and practical matter, how firms allocate resources so that they are in their first best use is a fundamental question. How firms build, augment, and modify their resource base over time is also of critical importance. In other words, there are important resource allocation functions that (neoclassical) economic theory ignores: namely, how does the non-market coordination inside, between, and among firms (necessary

for technology to get developed and deployed and for products and services to get produced) actually take place? Who performs that role? Economic theory yields poor answers. This is not surprising since there is nothing much that entrepreneurs and managers do in the economic theory of the market economy. Their role in the theory is often stripped out by the assumption of full information and the existence of a complete set of markets, even for contingent claims.²⁷

Economists have been very slow to recognize that much organizing is necessary before there are goods and services to exchange in markets. Adam Smith, in his famous pin-making example (Smith, 1776/1904, I.1.3), did not explain how the pin got invented and how the integration and coordination of non-traded pin sections (e.g., the wire, the head) took place inside the workshop to realize the fruits of specialization. Somewhat surprisingly, economists have not done much about this lacuna in the last 238 years.

It is also important to recognize that management is not just about specialization and the division of labor. It's also about ideation, co-creation, and co-ordination (orchestration). The integration of ideas and tasks to create or co-create innovative products and services is as important, if not more so, than specialization and the division of labor. This is not a recent development, but it is not yet adequately reflected in mainstream economic theory.

In short, the economic theory of markets needs to somehow recognize that a good deal of resource allocation takes place inside firms and between and among firms because of entrepreneurial and managerial decisions, activated by managerially designed systems. Yet there is no manager present in Adam Smith's pin-making example. The manager is scarcely present even in John Roberts' (2004) “modern firm.” When managers *are* present, the focus of modern economics and finance is the distribution, and less so the creation, of the spoils between managers and shareholders (Jensen, 2000). In reality, there had to be a manager in Smith's pin factory; it apparently did not register on Smith that to harness economies of specialization one needed managers.

Indeed, because of assumptions made, there is often no need for a manager in classical, neoclassi-

²⁶ Robbins (1932) was perhaps the first to frame the economic problem this way. It implicitly presupposes a knowledge of ends and means, which reduces the economic problem to one of technical efficiency and optimization.

²⁷ For a more complete statement about how management functions are obscured in economic theory, see Teece and Winter (1984).

cal, and modern economic theory. And if managers exist in economic theory, their primary motivation, it would seem from the very large literature on agency theory, is to steal from owners (Jensen & Meckling, 1976).²⁸ This raises the question of where the wealth inside firms comes from in the first place, something that agency theory doesn't address.

Logic and simple observation dictate that good managers play a most important role in value creation for the shareholder and for other constituencies such as employees. This is often forgotten in modern treatments of corporate governance and public policy. Agency theory has blotted out critical management functions, despite the important work of business historians, such as Chandler (1977), who highlight the positive role that managers play in the economic system.

Managers are essential to resource allocation and economic activity (and the theory of the firm). The price system has little relevance to the internal allocation of resources within firms because firms, for very good reasons, generally eschew the use of prices as the exclusive tool to determine the internal distribution of resources. The allocation process is instead orchestrated by managers.

One reason the price system is eschewed for internal resource allocation is that many assets and activities inside the firm are virtually impossible to price because they are highly specialized and idiosyncratic, their value is context-dependent, and there is little or no "liquidity" that would allow (internal) pricing to function as a good resource allocation mechanism. Clearly, the price system cannot effectuate the kind of tight coordination and co-evolution that dynamic capabilities require. Entrepreneurial managers, by contrast, can (and do) gather information, make assessments, and give directives so that non-priced assets are developed and deployed in value-enhancing ways. This is the orchestration function that the dynamic capabilities framework assigns to managers.

Hayek (1945) stressed the amazingly parsimonious powers of the price system to coordinate disparate economic activities around the globe. However, Hayek failed to note that management likewise achieves complex resource coordination, albeit inside the firm (and, in some cases, with the

firm's alliance partners). This managed coordination is an equally remarkable but very different kind of coordination to that of the price system, as it involves organizing the development and deployment of idiosyncratic, non-traded (and hence non-priced) assets without clear signals as to their value. Managerial orchestration is thus for the firm what prices are to the market in that they both enable coordination to achieve best-use resource allocation.

There are degrees of autonomy and automaticity attached to both types of coordination. Market coordination relies on the price system to generate signals that bring about responses from producers and consumers. Internal coordination is achieved by routines and buttressed by managers often making highly complex allocation decisions. Given its history of successes in the face of uncertainty, managerial orchestration certainly seems just as remarkable an allocation process as that which Hayek observed in the workings of the price system.

As explained in Teece (1980, 1982, 1986) and in Helfat and colleagues (2007, Chapter 2), managers, entrepreneurs, and innovators cannot just leave it up to a hypothetical market devoid of entrepreneurially managed firms to line up specific assets, develop new ones, and integrate them into a well-functioning innovation, production, and marketing system. The reason is that markets for high-specificity (idiosyncratic) assets generally don't exist, and if they do exist they are invariably "thin." To overcome this problem, managers collect information, sense opportunities, invest in capabilities, innovate, and transform. They become the instruments that help achieve the shrewd allocation of company resources. Markets and internal resource allocation (organized hierarchically inside the firm) are not only substitutes, as Coase (1937) implicitly claimed; they are also complements. Williamson seems to agree, noting that "the relation between competence and governance [is] both rival and complementary—more the latter than the former" (1999, p. 1106).

And even if transaction costs were zero and governance problems evaporated, learning, co-creation (Pitelis & Teece, 2009), and asset/resource orchestration functions would still need to be carried out. The entrepreneurially managed business firm is where this can be done.

The recognition that some form of organization (besides markets) is necessary in the economic system is hardly novel to management theorists. However, management theorists assume that organiza-

²⁸ See Bloom and Van Reenen (2007) for a notable exception to the virtual exclusion of firm-specific managerial practices from the economics literature.

tions are necessary and often don't explain why. The dynamic capabilities approach endeavors to explain why and, in doing so, creates insights into how the economy works that are often absent from economic theory.

Another standard problem given considerable attention in economic theory is the achievement of technical efficiency. Economists often assume that, if firms maximize profits, they will achieve efficiency. Leibenstein (1966), with his concept of x-inefficiency, which occurs when a firm operates above its cost curve, was perhaps the first economist (outside the Austrian School) to explicitly recognize that firms may not, in fact, achieve technological efficiency, and that the production function may therefore be different for different firms in the same industry. With the concept of x-inefficiency, Leibenstein made room for the possibility that managers might matter in economic theory, even if entrepreneurs didn't. However, Leibenstein's x-inefficiency theory, despite being cited occasionally, has not really been embraced by economists. It's an enigmatic dead end in the economics literature.

A very recent, welcome exception is Bloom and colleagues (2013), who declared in the *Quarterly Journal of Economics* that "management matters" (p. 40) based on a controlled study in which 14 Indian textile plants were taught a set of 38 well-known (in developed countries) management practices, resulting in a 17% increase in productivity in the first year.²⁹ The apparent reason for the firms' initial (avoidable) inefficiency was that the Indian managers had either not known about the superior practices or had been skeptical of what they had heard. This confirms basic Austrian School notions about imperfect information (and inaction) being ubiquitous in the economic system.

However, Bloom and colleagues (2013) did not characterize the economic role of the manager as being the orchestration of non-priced assets and of co-creation activity, as the dynamic capabilities framework implies. Rather, they focused on showing that management matters with respect to developing and using quite ordinary capabilities, which are amenable to transfer and testing in an experimental setting. In the dynamic capabilities frame-

work, that is merely the tip of the iceberg in terms of the ways that (entrepreneurial) management matters.

While not couched in the language of x-inefficiency, the dynamic capabilities framework implicitly accepts elements of that 50-year-old concept.³⁰ Leibenstein and others attributed x-inefficiency to a lack of competition, but the more fundamental reason is likely to be poor management, limited information, different histories, and weak ordinary and dynamic capabilities. Explanations along these lines have not garnered attention among economists (in trying to explain x-inefficiency) in Leibenstein's time or since.

The dynamic capabilities framework suggests a theory of the firm that not only accommodates firms with x-inefficiency (i.e., firms with weak ordinary capabilities, as evidenced by costs above the technically efficient level). It also accommodates firms that suffer from what might be called "d-ineffectiveness" (i.e., weak, ineffective dynamic capabilities). In fact, I posit that most firms are d-ineffective, because, at any point, many are likely to be offering a portfolio of products not ideally suited to current market needs.

Clearly, strategic management scholars have long recognized the problem of suboptimal management practices that, putting Leibenstein and Bloom et al. to one side, economic theory assumes away. As noted, a key tenet of the field of strategic management is that there is heterogeneity among firms. Not all firms will follow best practice, let alone generate and adapt new practices that outclass all others.

One way to remedy the shortfalls in economic theory described here is to embrace capability theory, or "capability economics," because it allows for (and helps explain) innovation and heterogeneity among firms. It also embraces strategy. While the Austrian School finds room for the entrepreneur, it doesn't have much room for the manager. In capability economics, there is a complementary place for the entrepreneur and the manager. Hence, capability theory takes Austrian economics to the next logical step.

²⁹ Google recently set out to test whether management matters at Google. They found that "even 'the smallest incremental increases in manager quality were quite powerful'" (Garvin, 2013, p. 77, citing Neal Patel, co-leader of Google's internal study).

³⁰ In particular, the dynamic capabilities framework recognizes that (1) not all firms are at the best practice frontier and (2) even those that have adopted best practice may be producing the "wrong" products (relative to current market requirements and technological opportunities) and they would do better to innovate.

CONCLUSIONS

Developments in the global economy have placed a premium on the ability of companies to become entrepreneurial and agile at home and abroad, requiring in turn that management formulate and follow good strategies and organize to allow and promote flexibility, learning, and, of course, innovation. This is the price of survival and growth in much of today's global economy. However, economic theory, if not management theory, is decades behind in recognizing these new realities.

The thesis advanced here is that good strategy, ownership of (or access to) VRIN resources, access to (but not necessarily ownership of) strong ordinary capabilities, scale (of some kind, in some circumstances), and strong dynamic capabilities augment the chances of an enterprise achieving long-term enterprise growth and survival. Put differently, what I referred to earlier as d-ineffective firms—those with weak dynamic capabilities—will have a short life, even if they don't suffer from x-inefficiency.

We have here the outlines of a (dynamic capabilities) theory of firms, one that encompasses how firms survive, grow, and prosper. As Gibbons (2005) noted, resources/capabilities approaches have "mouth-watering potential implications" for a formal theory of organizational development (2005, p. 202). The dynamic capabilities framework is distinguished from related approaches, such as Nonaka's (1994) knowledge spiral, O'Reilly and Tushman's (2004, 2008) ambidexterity, March's (1991) exploration and exploitation, or Argyris's (1976) double-loop learning in that it takes imitability and appropriability into account and shows how resources and strategy are intertwined with, and relate to, dynamic capabilities in establishing and sustaining competitive advantage.

With the dynamic capabilities framework, I would like to believe that we are indeed a few steps closer to a truly fundamental understanding of the origins of firm-level heterogeneity and the sources of enterprise-level value creation, capture, and growth. No other framework is as ambitious in its reach. As noted earlier, understanding the origins of long-term cash flow generation is the fundamental question in strategic management and also the deepest unanswered question in microeconomic and financial theory. Nor is it adequately addressed by organization theory. It is the question that directly and indirectly animates management theory

and investment decisions and motivates the quest for understanding enterprise survival and growth.

Teece and colleagues (1997) specified that rapid (technological) change gave salience to dynamic capabilities. In essence, rapid change requires firms to create and co-create, and to consistently assess and reassess whether they are fully taking advantage and contributing to the development of technological and market opportunities that provide what customers want, when they want it, and at desirable price points. In recent decades, the forces of globalization have amplified competition, exposing more markets to rapid change. This required new business models and made dynamic capabilities more salient. While globalization is by no means complete—in fact, semiglobalization (Ghemawat, 2003) is a more accurate characterization—globalization is unlikely to be reversed, short of a global economic calamity triggering a return to autarky.

In the dynamic capabilities framework, it is only what I call d-effective firms that are built to last. Ordinary capabilities are less salient and can often be outsourced. Efficiency alone is not enough for survival and growth. Capability theory provides the tools to allow (strategic) management theory to inform a deeper understanding of durable firm-level competitiveness, economic development, and the proper functioning of the economic system.

REFERENCES

- Adner, R., & Helfat, C. E. (2003). Corporate effects and dynamic managerial capabilities. *Strategic Management Journal*, 24(10), 1011–1025.
- AFP. (2013, February 12). Apple still has "magic," innovation, says CEO Cook. *afp.com*. Retrieved from <http://www.globalpost.com/dispatch/news/afp/130212/apple-still-has-magic-innovation-says-ceo-cook-0>
- Ambrosini, V., Bowman, C., & Collier, N. (2009). Dynamic capabilities: An exploration of how firms renew their resource base. *British Journal of Management*, 20(s1), S9–S24.
- Arend, R., & Bromiley, P. (2009). Assessing the dynamic capabilities view: Spare change, everyone? *Strategic Organization*, 7(1), 75–90.
- Argyris, C. (1976). Single-loop and double-loop models in research on decision making. *Administrative Science Quarterly*, 363–375.
- Augier, M., & Teece, D. J. (2008). Strategy as evolution with design: The foundations of dynamic capabilities.

- ties and the role of managers in the economic system. *Organization Studies*, 29(8–9), 1187–1208.
- Augier, M., & Teece, D. J. (2009). Dynamic capabilities and the role of managers in business strategy and economic performance. *Organization Science*, 20(2), 410–421.
- Bardolet, D., Lovallo, D., & Teece, D. J. (2014). Firm level dynamic capabilities and resource allocation: The empirical foundations of financial agility. Manuscript submitted for publication.
- Barney, J. B. (1986). Organizational culture: Can it be a source of sustained competitive advantage? *Academy of Management Review*, 11(3), 656–665.
- Barney, J. B. (1991). Firm resources and sustained competitive advantage. *Journal of Management*, 17(1), 99–120.
- Barney, J. B., & Clark, D. N. (2007). *Resource-based theory: Creating and sustaining competitive advantage*. Oxford, UK: Oxford University Press.
- Bartlett, C. A., & Ghoshal, S. (2002). *Managing across borders: The transnational solution* (2nd ed.). Cambridge, MA: Harvard Business School Press.
- Bartlett, C. A., & Rangan, U. S. (1986). *Kentucky Fried Chicken (Japan) Limited* (Case No. 9-387-043). Boston: Harvard Business School.
- Bloom, N., Eifert, B., Mahajan, A., McKenzie, D., & Roberts, J. (2013). Does management matter? Evidence from India. *Quarterly Journal of Economics*, 128(1), 1–51.
- Bloom, N., Genakos, C., Sadun, R., & Van Reenen, J. (2012). Management practices across firms and countries. *Academy of Management Perspectives*, 20(1), 12–33.
- Bloom, N., & Van Reenen, J. (2007). Measuring and explaining management practices across firms and countries. *Quarterly Journal of Economics*, 122(4), 1351–1408.
- Burrows, P. (2004, October 12). The seed of Apple's innovation. *Businessweek.com*. Retrieved from http://www.businessweek.com/print/bwdaily/dnflash/oct2004/nf20041012_4018_db083.htm
- Butler, D. A. (2006). *Distant victory: The battle of Jutland and the Allied triumph in the First World War*. Westport, CT: Praeger Security International.
- Chandler, A. D. (1977). *The visible hand*. Cambridge, MA: Harvard University Press.
- Chesbrough, H. W. (2003). *Open innovation: The new imperative for creating and profiting from technology*. Boston: Harvard Business School Press.
- China Economic Review. (2011, November 1). Original recipe: KFC remains far ahead of McDonald's in China. Retrieved from <http://www.chinaeconomicreview.com/content/original-recipe>
- Coase, R. A. (1937). The nature of the firm. *Económica*, 16(4), 386–405.
- Collis, D. J. (1994). Research note: How valuable are organizational capabilities? *Strategic Management Journal*, 15(S1), 143–152.
- Cyert, R. M., & March, J. G. (1963). *A behavioral theory of the firm*. Englewood Cliffs, NJ: Prentice Hall.
- D'Aveni, R. A. (1994). *Hypercompetition: Managing the dynamics of strategic maneuvering*. New York: Free Press.
- Danneels, E. (2002). The dynamics of product innovation and firm competences. *Strategic Management Journal*, 23(12), 1095–1121.
- Danneels, E. (2011). Trying to become a different type of company: Dynamic capability at Smith Corona. *Strategic Management Journal*, 32(1), 1–31.
- Di Stefano, G., Peteraf, M., & Verona, G. (2014). The organizational drivetrain: A road to integration of dynamic capabilities research. *Academy of Management Perspectives*, this issue.
- Dosi, G., Faillo, M., & Marengo, L. (2008). Organizational capabilities, patterns of knowledge accumulation and governance structures in business firms: An introduction. *Organization Studies*, 29(8–9), 1165–1185.
- Drnevich, P. L., & Kriauciunas, A. P. (2011). Clarifying the conditions and limits of the contributions of ordinary and dynamic capabilities to relative firm performance. *Strategic Management Journal*, 32(3), 254–279.
- Easterby-Smith, M., & Prieto, I. M. (2008). Dynamic capabilities and knowledge management: An integrative role for learning? *British Journal of Management*, 19(3), 235–249.
- Eisenhardt, K. M., & Martin, J. A. (2000). Dynamic capabilities: What are they? *Strategic Management Journal*, 21(10–11), 1105–1121.
- Garvin, D. A. (2013). How Google sold its engineers on management. *Harvard Business Review*, 91(12), 74–82.
- Gerstner, L. V. (2002). *Who says elephants can't dance? Inside IBM's historic turnaround*. New York: Harper-Business.
- Ghemawat, P. (2003). Semiglobalization and international business strategy. *Journal of International Business Studies*, 34(2), 138–152.
- Gibbons, R. (2005). Four formal(izable) theories of the firm? *Journal of Economic Behavior and Organization*, 58(2), 200–245.
- Gilbert, C. G. (2005). Unbundling the structure of inertia:

- Resource versus routine rigidity. *Academy of Management Journal*, 48(5), 741–763.
- Gratton, L., & Ghoshal, S. (2005). Beyond best practice. *MIT Sloan Management Review*, 46(3), 49–57.
- Greiner, L. E. (1998). Evolution and revolution as organizations grow. *Harvard Business Review*, 76(3), 55–60.
- Grindley, P. C., & Teece, D. J. (1997). Managing intellectual capital: Licensing and cross-licensing in electronics. *California Management Review*, 39(2), 8–41.
- Grove, A. S. (1996). *Only the paranoid survive: How to exploit the crisis points that challenge every company and career*. New York: Currency Doubleday.
- Hannan, M. T., & Freeman, J. (1977). The population ecology of organizations. *American Journal of Sociology*, 82(5), 929–964.
- Hayek, F. A. (1945). The use of knowledge in society. *American Economic Review*, 35(4), 519–530.
- Hayek, F. A. (1989). The pretense of knowledge. *American Economic Review*, 79(6), 3–7.
- Helfat, C. E., & Eisenhardt, K. M. (2004). Inter-temporal economies of scope, organizational modularity, and the dynamics of diversification. *Strategic Management Journal*, 25(13), 1217–1232.
- Helfat, C. E., Finkelstein, S., Mitchell, W., Peteraf, M. A., Singh, H., Teece, D. J., & Winter, S. G. (2007). *Dynamic capabilities: Understanding strategic change in organizations*. Oxford, UK: Blackwell.
- Helfat, C. E., & Peteraf, M. A. (2014). Managerial cognitive capabilities and the microfoundations of dynamic capabilities. *Strategic Management Journal*, in press. Retrieved from onlinelibrary.wiley.com/doi/10.1002/smj.2247
- Helfat, C. E., & Winter, S. G. (2011). Untangling dynamic and operational capabilities: Strategy for the (n)ever-changing world. *Strategic Management Journal*, 32(11), 1243–1250.
- Hoffman, F. (2004). What we can learn from Jackie Fisher. *Naval Institute Proceedings*, 130(4), 68–71.
- Jacobides, M. G., & Winter, S. G. (2012). Capabilities: Structure, agency, and evolution. *Organization Science*, 23(5), 1365–1381.
- Jensen, M. C. (2000). *A theory of the firm: Governance, residual claims, and organizational forms*. Cambridge, MA: Harvard University Press.
- Jensen, M. C., & Meckling, W. H. (1976). Theory of the firm: Managerial behavior, agency costs and ownership structure. *Journal of Financial Economics*, 3(4), 305–360.
- Kesmodel, D. (2011, December 30). Boeing examines supply chain for weak links. *WSJ.com*. Retrieved from <http://online.wsj.com/article/SB10001424052970204058404577111091095438300.html>
- Kirkpatrick, D. (1998, November 9). The second coming of Apple. *Fortune*. Retrieved from http://money.cnn.com/magazines/fortune/fortune_archive/1998/11/09/250834/index.htm
- Kleinbaum, A. M., & Stuart, T. E. (2014). Network responsiveness: The social structural microfoundations of dynamic capabilities. *Academy of Management Perspectives*, this issue.
- Lall, S., & Teubal, M. (1998). “Market-stimulating” technology policies in developing countries: A framework with examples from East Asia. *World Development*, 26(8), 1369–1385.
- Lashinsky, A. (2011, August 25). How Apple works: Inside the world’s biggest startup. *Fortune*. Retrieved from <http://tech.fortune.cnn.com/2011/08/25/how-apple-works-inside-the-worlds-biggest-startup>
- Leibenstein, H. (1966). Allocative efficiency vs. “X-inefficiency.” *American Economic Review*, 56(3), 392–419.
- Lippman, S. A., & Rumelt, R. P. (1982). Uncertain imitability: An analysis of inter-firm differences in efficiency under competition. *Bell Journal of Economics*, 13(2), 418–438.
- Lutz, B. (2011, June 11). Life lessons from the car guy. *WSJ.com*. Retrieved from <http://online.wsj.com/article/SB10001424052702304259304576375790237203556.html>
- March, J. G. (1991). Exploration and exploitation in organizational learning. *Organization Science*, 2(1), 71–87.
- March, J. G., & Simon, H. A. (1958). *Organizations*. New York: Wiley.
- Nelson, R. R., & Pack, H. (1999). The Asian miracle and modern economic growth. *Economic Journal*, 109(457), 416–436.
- Nelson, R. R., & Winter, S. G. (1982). *An evolutionary theory of economic change*. Cambridge, MA: Harvard University Press.
- Nonaka, I. (1994). A dynamic theory of organizational knowledge creation. *Organization Science*, 5(1), 14–37.
- Nonaka, I., & Toyama, R. (2007). Strategic management as distributed practical wisdom (phronesis). *Industrial and Corporate Change*, 16(3), 371–394.
- O’Reilly, C. A., & Tushman, M. L. (2004). The ambidextrous organization. *Harvard Business Review*, 82(4), 74–81.
- O’Reilly, C. A., & Tushman, M. (2008). Ambidexterity as a dynamic capability: Resolving the innovator’s di-

- lemma. *Research in Organizational Behavior*, 28, 185–206.
- Penrose, E. T. (1959). *The theory of the growth of the firm*. New York: John Wiley.
- Peteraf, M., Di Stefano, G., & Verona, G. (2013). The elephant in the room of dynamic capabilities: Bringing two diverging conversations together. *Strategic Management Journal*, 34(12), 1389–1410.
- Pisano, G., & Teece, D. J. (2007). How to capture value from innovation: Shaping intellectual property and industry architecture. *California Management Review* 50(1), 278–296.
- Pitelis, C. N., & Teece, D. J. (2009). The (new) nature and essence of the firm. *European Management Review*, 6(1), 5–15.
- Richardson, G. B. (1972). The organisation of industry. *Economic Journal*, 82(327), 883–896.
- Ries, E. (2011). *The lean startup: How today's entrepreneurs use continuous innovation to create radically successful businesses*. New York: Crown Business.
- Robbins, L. (1932). *An essay on the nature & significance of economic science*. London: Macmillan.
- Roberts, J. (2004). *The modern firm: Organizational design for performance and growth*. New York: Oxford University Press.
- Rose, G. (2013). Generation kill: A conversation with Stanley McChrystal. *Foreign Affairs*, 92(4), 2–8.
- Rosen, S. (1997). Austrian and neoclassical economics: Any gains from trade? *Journal of Economic Perspectives*, 11(4), 139–152.
- Rumelt, R. (2011). *Good strategy/bad strategy: The difference and why it matters*. New York: Crown Business.
- Schilke, O. (2014). Second-order dynamic capabilities: How do they matter? *Academy of Management Perspectives*, this issue.
- Sellers, P. (1993, April 19). Can this man save IBM? *Fortune*. Retrieved from http://archive.fortune.com/magazines/fortune/fortune_archive/1993/04/19/77748/index.htm
- Sirmon, D. G., Hitt, M. A., & Ireland, D. (2007). Managing firm resources in dynamic environments to create value: Looking inside the black box. *Academy of Management Review*, 32(1), 273–292.
- Smith, A. (1776/1904). In E. Cannan (Ed.), *An inquiry into the nature and causes of the wealth of nations*. London: Methuen and Co, Ltd. Retrieved from <http://www.econlib.org/library/Smith/smWN.html>
- Sutton, J. (1990). Explaining everything, explaining nothing? Game theoretic models in industrial economics. *European Economic Review*, 34(2), 505–512.
- Teece, D. J. (1980). Economies of scope and the scope of the enterprise. *Journal of Economic Behavior and Organization*, 1(3), 223–247.
- Teece, D. J. (1981). The market for know-how and the efficient international transfer of technology. *Annals of the American Academy of Political and Social Science*, 458(1), 81–96.
- Teece, D. J. (1982). Towards an economic theory of the multiproduct firm. *Journal of Economic Behavior and Organization*, 3(1), 39–63.
- Teece, D. J. (1986). Profiting from technological innovation. *Research Policy*, 15(6), 285–305.
- Teece, D. J. (1992). Competition, cooperation, and innovation: Organizational arrangements for regimes of rapid technological progress. *Journal of Economic Behavior and Organization*, 18(1), 1–25.
- Teece, D. J. (2000). *Managing intellectual capital: Organizational, strategic, and policy dimensions*. Oxford, UK: Oxford University Press.
- Teece, D. J. (2006). Reflections on profiting from innovation. *Research Policy*, 35(8), 1131–1146.
- Teece, D. J. (2007). Explicating dynamic capabilities: The nature and microfoundations of (sustainable) enterprise performance. *Strategic Management Journal*, 28(13), 1319–1350.
- Teece, D. J. (2009). *Dynamic capabilities and strategic management: Organizing for innovation and growth*. New York: Oxford University Press.
- Teece, D. J. (2011a). Achieving integration of the business school curriculum using the dynamic capabilities framework. *Journal of Management Development*, 30(5), 499–518.
- Teece, D. J. (2011b). Human capital, capabilities and the firm: Literati, numerati, and entrepreneurs in the 21st-century enterprise. In A. Burton-Jones & J.-C. Spender (Eds.), *The Oxford handbook of human capital* (pp. 527–562). Oxford, UK: Oxford University Press.
- Teece, D. J. (2012a). Next-generation competition: New concepts for understanding how innovation shapes competition and policy in the digital economy. *Journal of Law, Economics and Policy*, 9(1), 97–118.
- Teece, D. J. (2012b). Dynamic capabilities: Routines versus entrepreneurial action. *Journal of Management Studies*, 49(8), 1395–1401.
- Teece, D. J. (2014). A dynamic capabilities-based entrepreneurial theory of the multinational enterprise. *Journal of International Business Studies*, 45(1), 8–37.
- Teece, D. J., Pisano, G., & Shuen, A. (1997). Dynamic capabilities and strategic management. *Strategic Management Journal*, 18(7), 537–533.

- Tripsas, M., & Gavetti, G. (2000). Capabilities, cognition, and inertia: Evidence from digital imaging. *Strategic Management Journal*, 21(10–11), 1147–1161.
- Van Mieghem, J. A. (2008). *Operations strategy: Principles and practices*. Belmont, MA: Dynamic Ideas.
- Wernerfelt, B. (1984). A resource-based view of the firm. *Strategic Management Journal*, 5(2), 171–180.
- Williamson, O. E. (1975). *Markets and hierarchies*. New York: Free Press.
- Williamson, O. E. (1985). *The economic institutions of capitalism*. New York: Free Press.
- Williamson, O. E. (1999). Strategy research: Governance and competence perspectives. *Strategic Management Journal*, 20(12), 1087–1108.
- Winter, S. G. (2003). Understanding dynamic capabilities. *Strategic Management Journal*, 24(10), 991–995.
- Womack, J. P., Jones, D. T., & Roos, D. (1990). *The machine that changed the world: Based on the Massachusetts Institute of Technology 5-million-dollar 5-year study on the future of the automobile*. New York: Rawson Associates.
- Zahra, S. A., Sapienza, H. J., & Davidsson, P. (2006). Entrepreneurship and dynamic capabilities: A review, model and research agenda. *Journal of Management Studies*, 43(4), 917–955.



David J. Teece (teece@haas.berkeley.edu) is the Tusher Chair and director of the Tusher Center for Intellectual Capital at the Haas School of Business at the University of California, Berkeley. He is also chairman of the Berkeley Research Group and has published more than 20 books and 200 articles on the role of technology and capabilities in enterprise performance.

